

# Dunham Carbonate Classification: Depositional Fabrics & Grainstones

Comprehensive Textbook Study Guide & Complete Solution for DGMS 2019 Exam (Question 3)

Standard Reference: Sam Boggs Jr., *Principles of Sedimentology and Stratigraphy* (4th Edition, 2006)

## PART I

### Theoretical Foundation & Direct Textbook Analysis

Depositional textures, mud- vs. grain-support criteria, matrix winnowing, and the Embry & Klovan reef expansion

## 1. THE PHILOSOPHY OF CARBONATE CLASSIFICATION: TEXTURE VS. COMPOSITION

In carbonate petrology, two seminal classification schemes predominate: **Robert L. Folk's (1959, 1962)** compositional classification (based on allochems and matrix type: micrite vs. sparite) and **Robert J. Dunham's (1962)** textural classification.

*"Dunham (1962) devised a classification of limestones based on the depositional texture of the rocks rather than the identity of different kinds of carbonate grains. Dunham's classification is based on whether the components of the rock were bound together during deposition (boundstone) or were not bound together. Rocks whose components were not bound together are divided on the basis of whether the rock is matrix-supported (mudstone, wackestone) or grain-supported (packstone, grainstone). The presence or absence of lime mud (< 20 µm) serves as the primary indicator of depositional energy."*

— Sam Boggs Jr., *Principles of Sedimentology and Stratigraphy* (4th Ed., 2006), p. 188

Dunham's system is universally favored for hand-specimen, core, and thin-section description because it directly translates into **hydrodynamic energy conditions at the time of deposition**:

- **Presence of Lime Mud (Micrite):** Indicates calm, low-energy waters (lagoons, protected bays, deep marine below wave base) where fine carbonate mud can settle from suspension.
- **Absence of Lime Mud (Washed Clean):** Indicates turbulent, high-energy settings (shoals, barrier bars, beaches, tidal passes) where waves and currents vigorously winnow away the fine mud, leaving only a grain-supported framework that is later cemented by sparry calcite cement.

## 2. DUNHAM'S (1962) CORE CLASSES & DISTINCTIONS

Dunham established four primary classes for non-bound carbonate rocks based on the ratio of mud to grains and framework support:

- **Mudstone:** Mud-supported fabric containing **less than 10% grains** (>90% lime mud matrix). Reflects ultra-quiet, stagnant or sheltered water.
- **Wackestone:** Mud-supported fabric containing **more than 10% grains**, but grains do not touch to support one another; they "float" in a matrix of lime mud.
- **Packstone:** **Grain-supported framework** containing **lime mud matrix** in the interstitial spaces between grains. Represents moderate energy or rapid dumping where mud was not completely winnowed.
- **Grainstone:** **Grain-supported framework** that contains **almost no lime mud matrix** (<1% mud). Interstitial voids are completely free of mud and are filled by sparry calcite cement. Reflects persistent, high-energy wave or current winnowing.
- **Boundstone:** Carbonate components were bound together organically during deposition by colonial organisms (corals, stromatoporoids, encrusting algae, bryozoans).

**Table 6.3** Classification of limestones according to depositional textures

| A                                                        |                      |           |                 |                                  | DEPOSITIONAL TEXTURE RECOGNIZABLE                                                                                                                                                                                                                                        |  | DEPOSITIONAL TEXTURE NOT RECOGNIZABLE                                                                                     |  |  |
|----------------------------------------------------------|----------------------|-----------|-----------------|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|---------------------------------------------------------------------------------------------------------------------------|--|--|
| Original components not bound together during deposition |                      |           |                 |                                  | Original components were bound together during deposition . . . as shown by intergrown skeletal matter, lamination contrary to gravity, or sediment-floored cavities that are roofed over by organic or questionably organic matter and are too large to be interstices. |  | CRYSTALLINE CARBONATE<br><br>(Subdivide according to classifications designed to bear on physical texture or diagenesis.) |  |  |
| Contains mud (particles of clay and fine silt size)      |                      |           | Grain-supported | Lacks mud and is grain-supported |                                                                                                                                                                                                                                                                          |  |                                                                                                                           |  |  |
| Mud-supported                                            |                      |           |                 |                                  |                                                                                                                                                                                                                                                                          |  |                                                                                                                           |  |  |
| Less than 10% grains                                     | More than 10% grains |           |                 |                                  |                                                                                                                                                                                                                                                                          |  |                                                                                                                           |  |  |
| MUDSTONE                                                 | WACKSTONE            | PACKSTONE | GRAINSTONE      | BOUNDSTONE                       |                                                                                                                                                                                                                                                                          |  |                                                                                                                           |  |  |

| B                                                           |                         |                 |             |            |                                   | ALLOCHTHONOUS LIMESTONE                                 |                                           |                                    | AUTOCHTHONOUS LIMESTONE          |             |  |  |
|-------------------------------------------------------------|-------------------------|-----------------|-------------|------------|-----------------------------------|---------------------------------------------------------|-------------------------------------------|------------------------------------|----------------------------------|-------------|--|--|
| Original components not organically bound during deposition |                         |                 |             |            |                                   | Original components organically bound during deposition |                                           |                                    |                                  |             |  |  |
| Less than 10% >2 mm components                              |                         |                 |             |            | Greater than 10% >2 mm components |                                                         | By organisms that build a rigid framework | By organisms that encrust and bind | By organisms that act as baffles |             |  |  |
| Contains lime mud (<0.03 mm)                                |                         |                 | No lime mud |            | Matrix-supported                  | >2 mm component-supported                               |                                           |                                    |                                  |             |  |  |
| Mud-supported                                               |                         |                 |             |            |                                   |                                                         |                                           |                                    |                                  |             |  |  |
| Less than 10% grains >0.03 mm <2 mm                         | Greater than 10% grains | Grain-supported |             |            |                                   |                                                         |                                           |                                    |                                  |             |  |  |
| MUDSTONE                                                    | WACKSTONE               | PACKSTONE       | GRAINSTONE  | FLOATSTONE | RUDSTONE                          | FRAMESTONE                                              | BINDSTONE                                 | BOUNDSTONE                         |                                  | BAFFLESTONE |  |  |

Source: A. After Dunham, R. J., 1962, Classification of carbonate rocks according to depositional textures, in Ham, W. E., ed., Classification of carbonate rocks, Am. Assoc. Petroleum Geologists Mem. 1, Table 1, p. 117, reprinted by permission of AAPG, Tulsa, Okla. B. after Dunham, R. J., 1962, as modified by Embry, E. F., III and J. E. Klovan, 1972, Absolute water depth limits of late Devonian paleoecological zones, Geol. Rundschau, v. 61, Fig. 5, p. 676, reprinted by permission.

**Table 6.3 (Boggs, *Principles of Sedimentology and Stratigraphy*, 2006, p. 189):** Table 6.3A: The original Dunham (1962) classification of carbonate rocks according to depositional texture. Notice the explicit definition of **Grainstone**: "Grain-supported, Lacks mud". Table 6.3B: The Embry and Klovan (1971) modification expanding boundstones and coarse (>2 mm) carbonate gravels.

### 3. THE EMBRY & KLOVAN (1971) REEF & MEGAGRAIN EXPANSION

Embry and Klovan (1971) expanded Dunham's system to better classify organic reef rocks and coarse bioclastic carbonates containing particles larger than 2 mm (Boggs, 2006, pp. 188–189):

- **Autochthonous Reef Rocks (Bound in Life):**

- **Framestone:** Supported by a rigid organic framework (e.g., massive branching corals, stromatoporoids).
- **Bindstone:** Supported by tabular or encrusting organisms that bind and laminate sediment (e.g., microbial mats, encrusting red algae).
- **Bafflestone:** Supported by stalk-like organisms that baffle current flow and trap sediment (e.g., upright bryozoans, sea grasses).

- **Allochthonous Coarse Carbonates (>10% grains > 2 mm):**

- **Floatstone:** Matrix-supported carbonate gravel (grains > 2 mm float in a fine matrix).
- **Rudstone:** Grain-supported carbonate gravel (grains > 2 mm form the self-supporting framework).

## PART II

### DGMS 2019 Examination: Question & Worked Solution

Rigorous option-by-option critical analysis and definitive textbook-grounded answer

#### ORIGINAL QUESTION — DGMS 2019 (JUNIOR SCIENTIFIC OFFICER)

**Q.3. A carbonate rock that contains almost no mud and shows grain-supported framework is named as**

(a) **Grainstone**

(b) **Packstone**

(c) **Floatstone**

(d) **Bindstone**

#### Correct Answer: (a) Grainstone



By Dunham's (1962) universal classification, a carbonate rock possessing a self-supporting grain framework and devoid of lime mud (matrix-free) is definitively named a Grainstone.

#### OPTION-BY-OPTION CRITICAL ANALYSIS

##### ✓ (a) Grainstone

**CORRECT**

**Why it is correct:** As defined in Table 6.3A of Boggs (2006, p. 189) and Dunham (1962), a **grainstone** is a grain-supported carbonate rock that "lacks mud" (contains almost no micrite). The intergranular pore spaces are typically filled by clear, sparry calcite cement that precipitated during diagenesis after deposition.

##### ✗ (b) Packstone

**INCORRECT**

**Why it is incorrect:** While a packstone is grain-supported, it is defined by the **presence of lime mud** in its interstitial pore spaces ("grain-supported, contains mud"; Boggs, p. 189). Because the question explicitly specifies "contains almost no mud", packstone is eliminated.

**X (c) Floatstone**

**INCORRECT**

**Why it is incorrect:** In the Embry & Klovan (1971) classification, a floatstone is a coarse carbonate rock containing >10% grains larger than 2 mm, but it is **matrix-supported** (the large clasts float in a finer matrix). It is neither mud-free nor grain-supported.

**X (d) Bindstone**

**INCORRECT**

**Why it is incorrect:** A bindstone is an autochthonous reef carbonate where organisms (such as tabular encrusting red algae or cyanobacterial mats) bound the sediment together during life. The question describes an un-bound clastic depositional framework.

PART  
III

Synthesis Reference Tables & High-Yield Practice Questions

Quick-reference diagnostic tables and 8 curated exam questions for DGMS, UPSC GSI, and IIT-JAM

SYNTHESIS TABLES: DUNHAM & EMBRY-KLOVAN FRAMEWORK

Table 1: Dunham (1962) Classification of Carbonates

| Class      | Support Fabric            | Lime Mud Content         | Depositional Energy | Typical Environment                      |
|------------|---------------------------|--------------------------|---------------------|------------------------------------------|
| Mudstone   | Mud-supported             | > 90% mud (< 10% grains) | Very Low            | Protected lagoons, deep basinal marine   |
| Wackestone | Mud-supported             | < 90% mud (> 10% grains) | Low to Moderate     | Shallow shelf lagoon, quiet embayment    |
| Packstone  | Grain-supported           | Contains mud matrix      | Moderate            | Sub-wave-base flank, tempestite dump     |
| Grainstone | Grain-supported           | Lacks mud (< 1%)         | High (winnowed)     | Ooid shoals, tidal inlets, beach barrier |
| Boundstone | Organically bound in life | Variable                 | High to Moderate    | Biogenic reef core, organic framework    |

**Table 2: Folk (1959/1962) vs. Dunham (1962) Equivalence**

| Dunham Class | Primary Characteristic                  | Equivalent Folk (1959) Term (Ooid example) | Equivalent Folk (1959) Term (Bioclast example) |
|--------------|-----------------------------------------|--------------------------------------------|------------------------------------------------|
| Grainstone   | Grain-supported, spar cement, mud-free  | Oosparite                                  | Biosparite                                     |
| Packstone    | Grain-supported, micrite matrix present | Packed Oomicrite                           | Packed Biomicrite                              |
| Wackestone   | Mud-supported, > 10% grains             | Sparse Oomicrite                           | Sparse Biomicrite                              |
| Mudstone     | Mud-supported, < 10% grains             | Micrite                                    | Micrite / Fossiliferous Micrite                |
| Boundstone   | Organically bound framework             | Biolithite                                 | Biolithite                                     |

**PROBABLE HIGH-YIELD EXAM QUESTIONS (UPSC GSI, DGMS, IIT-JAM & GATE)**

**Q1. A carbonate rock consisting of 60% ooids and 40% sparry calcite cement with no lime mud is classified in Folk's and Dunham's schemes respectively as:**

- (a) Oomicrite / Packstone
- (b) Oosparite / Grainstone
- (c) Oosparite / Wackestone
- (d) Oomicrite / Grainstone

→ **Correct Answer: (b) Oosparite / Grainstone**

Explanation: In Folk's system, ooids + spar = Oosparite. In Dunham's system, grain-supported + mud-free = Grainstone.

**Q2. In the Embry & Klovan (1971) classification, a reef limestone supported by organisms that acted as sediment baffles during life is termed:**

- (a) Bindstone    (b) Bafflestone    (c) Framestone    (d) Rudstone

→ **Correct Answer: (b) Bafflestone**

Explanation: Bafflestones are formed by upright, stalked organisms (e.g., bryozoans, crinoids, sea grasses) that trap and baffle current-borne sediment.

**Q3. A limestone consisting of 30% fossil shells floating in a lime mud matrix is named in Dunham's classification as:**

- (a) Mudstone    (b) Wackestone    (c) Packstone    (d) Grainstone

→ **Correct Answer: (b) Wackestone**

Explanation: A wackestone is mud-supported and contains >10% grains (here 30%). If it contained <10% grains, it would be a mudstone.

**Q4. How does a Rudstone differ from a Floatstone in the Embry & Klovan scheme?**

- (a) Rudstone is matrix-supported; Floatstone is grain-supported
- (b) Rudstone is grain-supported; Floatstone is matrix-supported
- (c) Rudstone contains < 2 mm grains; Floatstone contains > 2 mm grains
- (d) Rudstone is organically bound; Floatstone is unbound

→ **Correct Answer: (b) Rudstone is grain-supported; Floatstone is matrix-supported**

Explanation: Both rocks have >10% gravel-size (>2 mm) carbonate particles, but Rudstones have a grain-supported framework, whereas Floatstones are matrix-supported.

**Q5. The boundary grain size separating lime mud (micrite) from sand-sized carbonate allochems in Dunham's classification is:**

- (a) 2  $\mu\text{m}$     (b) 20  $\mu\text{m}$     (c) 63  $\mu\text{m}$     (d) 2 mm

→ **Correct Answer: (b) 20  $\mu\text{m}$**

Explanation: Dunham (1962) defined lime mud as carbonate particles finer than 20  $\mu\text{m}$ , encompassing micrite (<4  $\mu\text{m}$ ) and microspar (4–20  $\mu\text{m}$ ).

**Q6. The presence of a Grainstone facies in a stratigraphic succession indicates:**

- (a) Deep marine euxinic basin
- (b) High-energy agitated shallow water shoal or beach
- (c) Stagnant restricted coastal lagoon
- (d) Glaciomarine dropstone setting

→ **Correct Answer: (b) High-energy agitated shallow water shoal or beach**

Explanation: Grainstones require strong, persistent wave or tidal currents to continuously wash and winnow away fine lime mud.

**Q7. An encrusting stromatopore or red algal framework that binds sediment is categorized as a:**

- (a) Bafflestone    (b) Bindstone    (c) Floatstone    (d) Grainstone

→ **Correct Answer: (b) Bindstone**

Explanation: Organisms that encrust and bind sheet-like organic crusts form bindstones.

**Q8. Packstone differs from Grainstone primarily in:**

- (a) Grain size of the allochems
- (b) Presence versus absence of lime mud in the matrix
- (c) Mineralogy (calcite vs aragonite)
- (d) Degree of dolomitization

→ **Correct Answer: (b) Presence versus absence of lime mud in the matrix**

Explanation: Both are grain-supported; packstones contain lime mud matrix, whereas grainstones are completely winnowed clean of mud.

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Source Exam: DGMS 2019 (Junior Scientific Officer), Question 3 · Subject: Carbonate Sedimentology

Textbook Authority: Sam Boggs Jr., *Principles of Sedimentology and Stratigraphy* (4th Ed., 2006), Chapter 6, pp. 187–192